

Do democracies actually work?

If a majority of citizens want a policy, does it become law?

Most people assume "yes, eventually." That's the whole pitch of representative democracy: preferences in, policy out, some friction in the middle.

I wanted to know how much friction. So I built an agent-based model of four democratic legislatures (pure parliamentary, pure presidential, and the two semi-presidential hybrids) and ran each one 200 times across four political scenarios: baseline, fragmented, polarised, small.

That's 3,200 simulated legislative sessions. The code is open source at github.com/tofuadmiral/institutional-representation-abm and there's a live demo at institutional-representation-abm.streamlit.app where you can move every dial yourself.

Three things surprised me.

1. Parliaments don't fail gracefully. They collapse.

Under fragmentation (five small parties, no natural majority) a pure parliamentary system passes **0.05%** of bills.

Not fewer bills. Not slower bills. Effectively zero. No coalition holds together long enough to govern, and every attempt triggers another confidence vote.

Political scientists have been arguing versions of this about Italy and Israel for decades. In the model it shows up cleanly in seed-averaged data with bootstrap confidence intervals (which was a relief, nothing worse than a headline finding that dies at $n=200$).

2. The thing that kills parliaments also saves them

I have a mechanism ablation called `no_discipline` that lets legislators ignore their party whip. Run the fragmented parliament without discipline and passage jumps from 0.05% to **46.7%**.

The whip, the thing that makes MPs look like puppets, is what lets a fragmented parliament aggregate votes at all. Take it out and you get chaos. Leave it in and you get governance.

But the sign flips. Under polarisation (two ideologically distant parties, wide policy proposals), the whip is what hurts: enforcing discipline costs the parliament 66 percentage points of passage, because every bill becomes a partisan litmus test.

So whether discipline helps depends on which problem you have: too many factions, or two locked factions. Most reform debates quietly assume one of those worlds and prescribe for it.

The model says check which world you're in first.

The same ablation across all four institutions gives a clean monotone ordering:

Institution	Passage @ fragmented	Δ when discipline removed
Parliamentary	0.001	+0.466
Premier-presidential	0.013	+0.312
President-parliamentary	0.086	+0.233
Republican	0.448	-0.242

The more an institution depends on forming a majority government, the more it depends on the whip. Presidential systems decouple the executive from legislative coalitions, so they don't need it (and are actively hurt by it).

3. There is no best system, just a dial

I added a metric called the **policy representation gap**: the L2 distance between the bills that pass and the median preference of the constituencies they're supposed to represent.

Under polarisation:

Institution	Passage rate	Representation gap
Parliamentary	88.2%	+0.037
Premier-presidential	72.5%	-0.012
President-parliamentary	73.0%	-0.010
Republican	21.8%	-0.530

The parliamentary system passes four times as many bills, but they drift away from what voters want. The presidential system passes far fewer, and the ones that survive the veto sit much closer to the public's median.

This is the deepest finding in the project. Throughput and fidelity live on the same spectrum. You cannot max both. Every real democracy is a point on this curve, and "more democratic" turns out to be a choice about which axis you're willing to give up.

So, do democracies work?

Depends what "work" means.

- *Passes what the majority wants, fast*: parliaments, right up until they collapse catastrophically.
- *Only passes laws that match the public*: presidential systems, at the cost of long stretches where nothing passes at all.
- *Survives whichever crisis shows up*: the hybrids are doing something interesting in the middle. They lose on both metrics individually but lose less when the scenario shifts. There's a real case for designs whose worst case is mediocre rather than catastrophic.

One caveat: the model is generative, in the Kollman-Miller-Page and Laver-Sergenti tradition. It isolates structural tradeoffs, it doesn't forecast what Italy does next year. But the tradeoffs it isolates are sharp, reproducible (every result is seed-deterministic and CI-gated), and uncomfortable for anyone whose theory of democracy has only one number in it.

Model, paper, sensitivity analyses, and the interactive demo are all here:

github.com/tofuadmiral/institutional-representation-abm. Go break a parliament, it's weirdly fun.